

# Feature Spec: Invoices & Payments

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## Note to Developer:

1. **Stripe Test Account** — Use your own Stripe test account for the Stripe integration. Create one at [dashboard.stripe.com](https://dashboard.stripe.com) if you don't have one. Use test mode keys only — never real keys in development.
2. **Loom Recording** — When you finish this feature, record a Loom video walking through everything you built. Show the full flow: creating an appointment, the auto-generated invoice, recording payments, refunds, manual invoice creation, filters, and the balance displays on customer and appointment tabs. This video is mandatory before the PR can be reviewed.

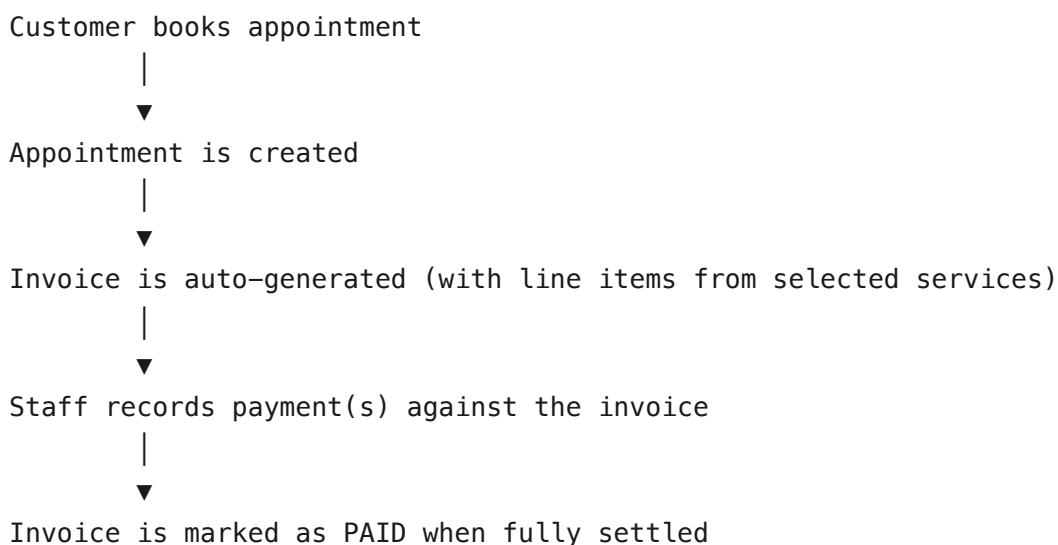
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## 1. Overview

### What Is This?

This feature adds the billing and payment layer to the CRM. Right now we have appointments, customers, doctors, and services — but no way to charge for anything. This feature connects money to appointments.

### How It Fits



### What We Are Building

Three new database tables:

- **Invoices** — the bill
- **Invoice Items** — the line items on the bill (one per service)
- **Payments** — the money received against the bill

Plus the UI to manage all of this in the portal.

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## 2. Database Tables

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### 2.1 Invoices Table

This is the main billing record. One invoice per appointment (auto-generated), or created manually by staff for standalone charges.

| Column          | Type    | Required | Notes                                                       |
|-----------------|---------|----------|-------------------------------------------------------------|
| id              | UUID    | Yes      | Primary key                                                 |
| clinic_id       | UUID    | Yes      | Tenant isolation — EVERY query must filter by this          |
| appointment_id  | UUID    | No       | Null for manual invoices (not tied to an appointment)       |
| customer_id     | UUID    | Yes      | FK to customers table                                       |
| invoice_number  | VARCHAR | Yes      | Unique per clinic, e.g. "INV-2026-0042"                     |
| status          | ENUM    | Yes      | See status lifecycle below                                  |
| subtotal        | INTEGER | Yes      | Sum of all line items (in cents/pence)                      |
| discount_type   | ENUM    | No       | PERCENTAGE or FIXED (invoice-level discount)                |
| discount_value  | INTEGER | No       | Discount amount (percentage value or fixed amount in cents) |
| discount_amount | INTEGER | Yes      | Calculated discount in cents (default 0)                    |
| tax_rate        | DECIMAL | Yes      | Tax percentage at time of invoice creation (snapshot)       |
| tax_label       | VARCHAR | Yes      | "VAT", "GST", "Tax", etc. (snapshot from clinic settings)   |
| tax_amount      | INTEGER | Yes      | Calculated tax in cents                                     |
| total_amount    | INTEGER | Yes      | subtotal - discount_amount + tax_amount                     |
| amount_paid     | INTEGER | Yes      | Sum of all successful payments (default 0)                  |
| amount_due      | INTEGER | Yes      | total_amount - amount_paid                                  |
| due_date        | DATE    | Yes      | When payment is expected                                    |
| notes           | TEXT    | No       | Internal notes from staff                                   |

|            |           |     |      |
|------------|-----------|-----|------|
| created_at | TIMESTAMP | Yes | Auto |
| updated_at | TIMESTAMP | Yes | Auto |

**Important:** All money fields are **integers in cents/pence** (e.g., 25000 = £250.00). Never use floats for money.

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## 2.2 Invoice Items Table

Each row represents one service (line item) on an invoice. These are **snapshots** — if the service price changes later, old invoices are not affected.

| Column          | Type      | Required | Notes                                             |
|-----------------|-----------|----------|---------------------------------------------------|
| id              | UUID      | Yes      | Primary key                                       |
| clinic_id       | UUID      | Yes      | Tenant isolation                                  |
| invoice_id      | UUID      | Yes      | FK to invoices table                              |
| service_id      | UUID      | No       | FK to services table (null for custom line items) |
| description     | VARCHAR   | Yes      | Service name at time of invoicing (snapshot)      |
| quantity        | INTEGER   | Yes      | Default 1                                         |
| unit_price      | INTEGER   | Yes      | Price per unit in cents (snapshot from service)   |
| discount_type   | ENUM      | No       | PERCENTAGE or FIXED (line-item discount)          |
| discount_value  | INTEGER   | No       | Discount amount                                   |
| discount_amount | INTEGER   | Yes      | Calculated discount in cents (default 0)          |
| line_total      | INTEGER   | Yes      | (quantity × unit_price) - discount_amount         |
| created_at      | TIMESTAMP | Yes      | Auto                                              |

**Why snapshot the description and price?** Because services change. If Botox goes from £250 to £300 next month, Sarah's old invoice should still show £250. The invoice item is a frozen record of what was charged at that moment.

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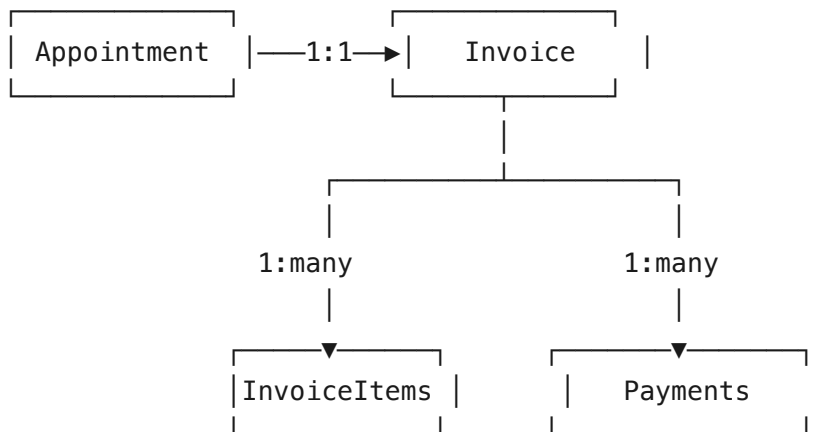
## 2.3 Payments Table

Each row is a single payment transaction against an invoice. An invoice can have many payments (deposits, split payments, etc.).

| Column    | Type | Required | Notes            |
|-----------|------|----------|------------------|
| id        | UUID | Yes      | Primary key      |
| clinic_id | UUID | Yes      | Tenant isolation |

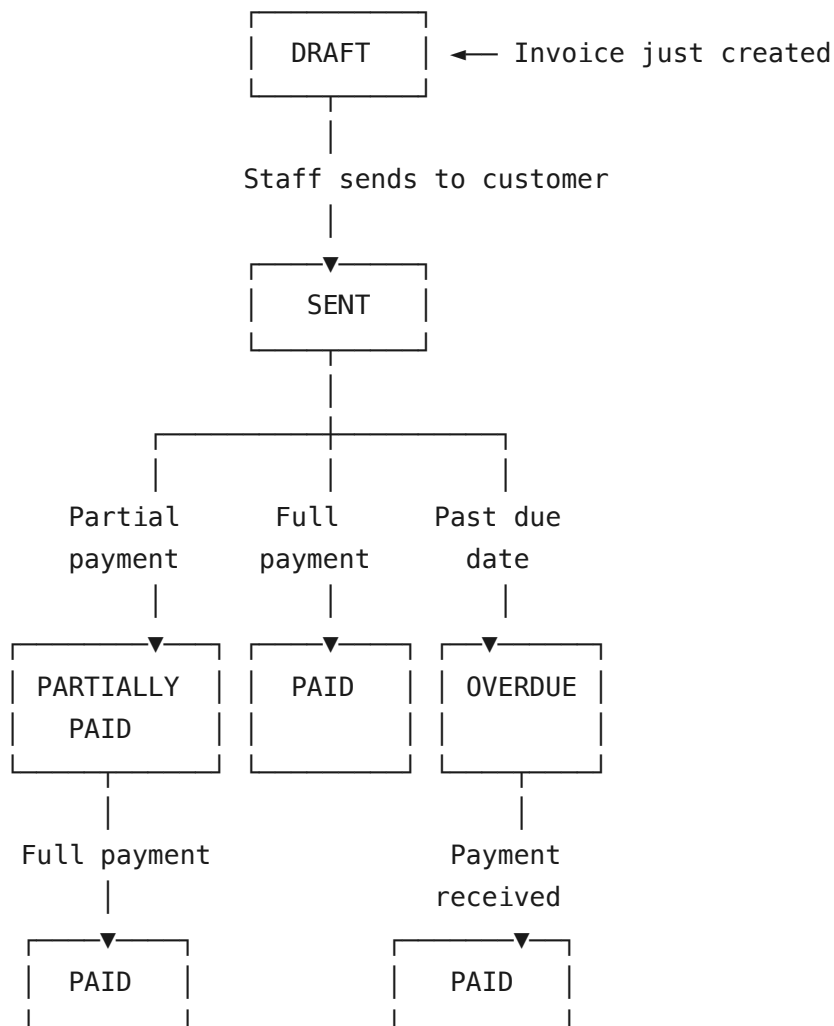
|                        |           |     |                                                              |
|------------------------|-----------|-----|--------------------------------------------------------------|
| invoice_id             | UUID      | Yes | FK to invoices table                                         |
| type                   | ENUM      | Yes | PAYMENT or REFUND                                            |
| amount                 | INTEGER   | Yes | Amount in cents (positive for payment, negative for refund)  |
| payment_method         | ENUM      | Yes | CASH, CARD, BANK_TRANSFER                                    |
| payment_gateway        | ENUM      | Yes | MANUAL or STRIPE                                             |
| gateway_transaction_id | VARCHAR   | No  | Stripe payment intent ID (null for manual/cash)              |
| linked_payment_id      | UUID      | No  | For refunds — references the original payment being refunded |
| status                 | ENUM      | Yes | SUCCEEDED, FAILED, REFUNDED                                  |
| notes                  | TEXT      | No  | "Paid at reception", "Deposit before appointment", etc.      |
| paid_at                | TIMESTAMP | Yes | When the payment was made (can be backdated)                 |
| created_at             | TIMESTAMP | Yes | Auto                                                         |

### 3. Relationships

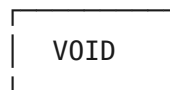


- **Appointment → Invoice:** One appointment generates one invoice. But an invoice can also exist without an appointment (manual invoices).
- **Invoice → Invoice Items:** One invoice has many line items. Each item = one service or custom charge.
- **Invoice → Payments:** One invoice can have many payments. Each payment is a separate transaction (deposit, final payment, refund, etc.).
- **Payment → Payment (self-reference):** A refund payment links back to the original payment via `linked_payment_id`.

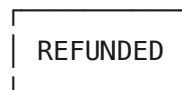
## 4. Invoice Status Lifecycle



At any point before PAID:



After PAID:



| Status                | When                                      | What Happens                       |
|-----------------------|-------------------------------------------|------------------------------------|
| <b>DRAFT</b>          | Invoice just created (auto or manual)     | Not visible to customer yet        |
| <b>SENT</b>           | Staff sends/shares with customer          | Customer can now see it            |
| <b>PARTIALLY_PAID</b> | Some payment received but not full amount | amount_paid > 0 but < total_amount |
| <b>PAID</b>           | Full amount received                      | amount_due = 0                     |
| <b>OVERDUE</b>        | Past due_date and not fully paid          | Set by automated cron job daily    |

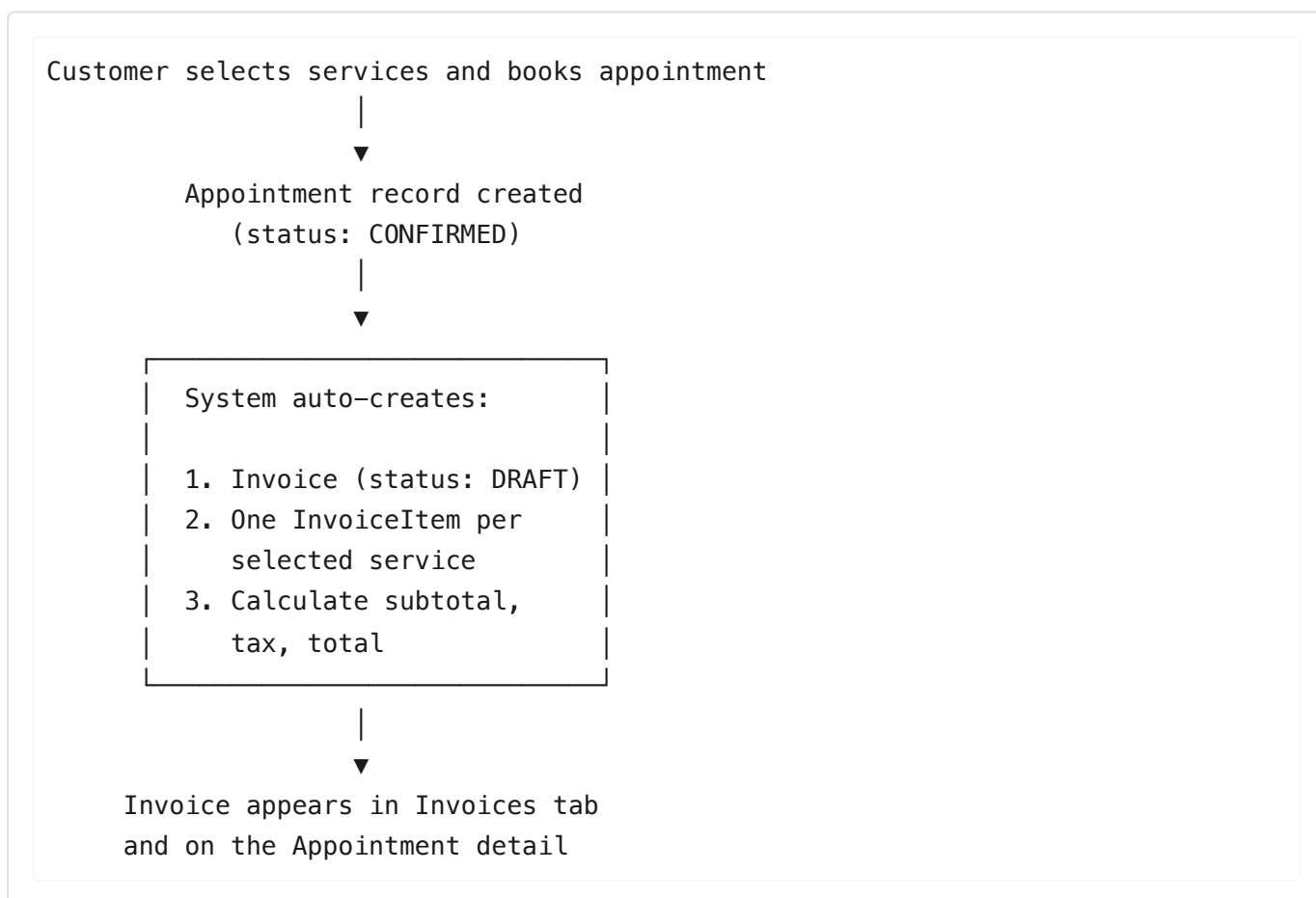
|                 |                                  |                                   |
|-----------------|----------------------------------|-----------------------------------|
| <b>VOID</b>     | Cancelled before any payment     | No money movement, invoice killed |
| <b>REFUNDED</b> | Full refund issued after payment | Money returned to customer        |

## 5. Payment Status

| Status           | Meaning                                      |
|------------------|----------------------------------------------|
| <b>SUCCEEDED</b> | Payment went through successfully            |
| <b>FAILED</b>    | Payment attempt failed (card declined, etc.) |
| <b>REFUNDED</b>  | This payment was later refunded              |

## 6. Flows

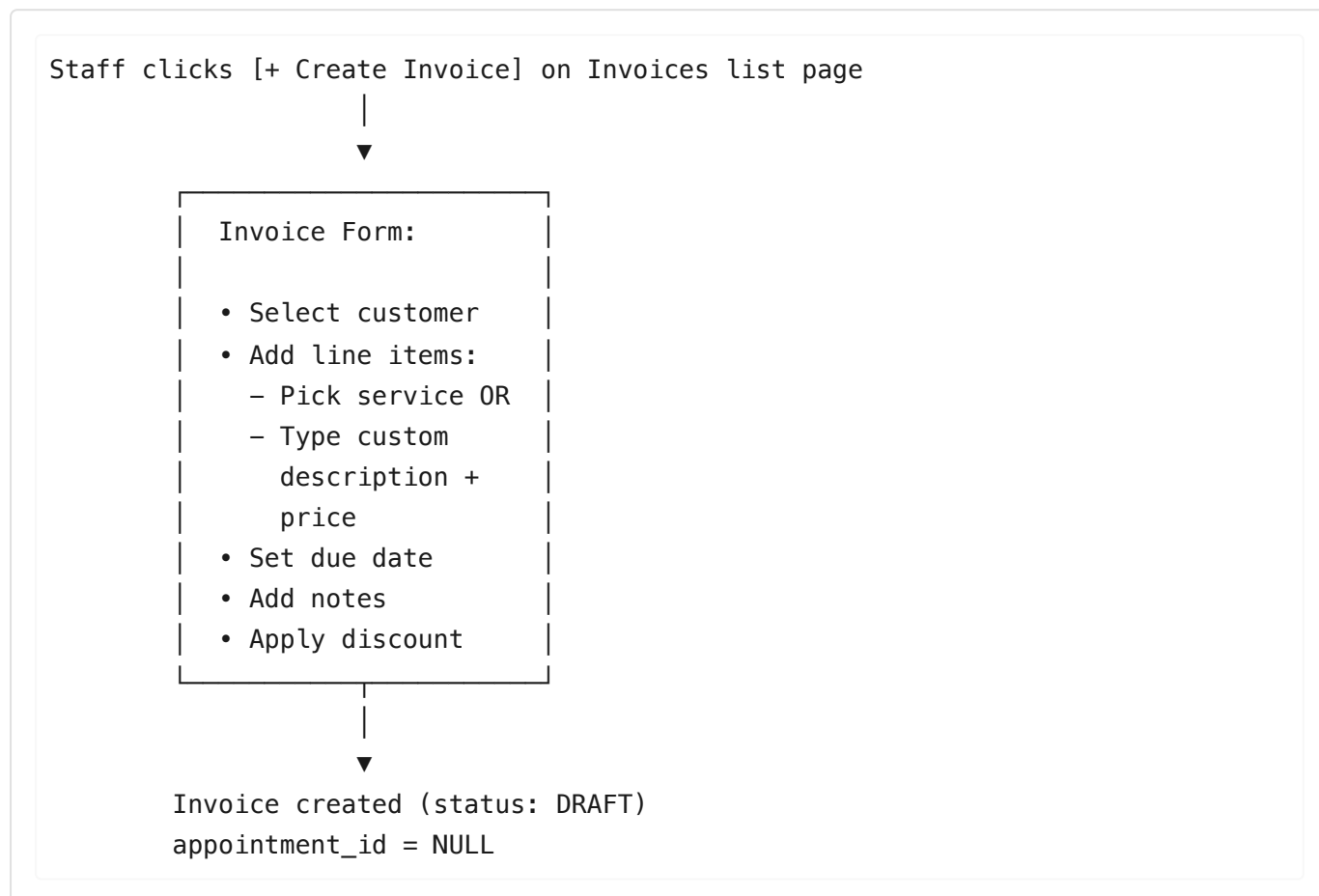
### 6.1 Auto Invoice Creation (On Appointment Booking)



**Where do the prices come from?** From the services table. When the invoice is created, it snapshots the current price of each service into the invoice item's `unit_price` and `description` . After this point, the invoice item is independent of the service record.

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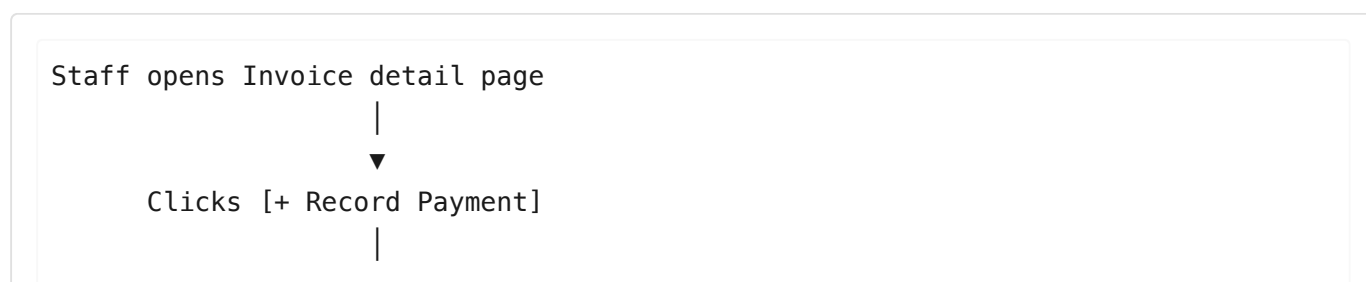
## 6.2 Manual Invoice Creation (No Appointment)

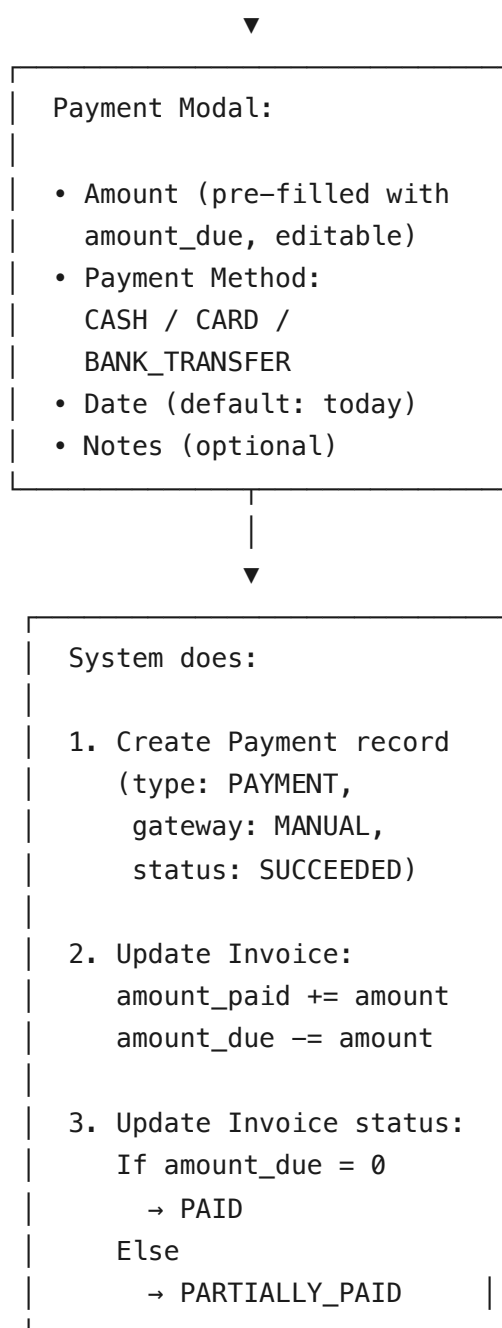


### Use cases for manual invoices:

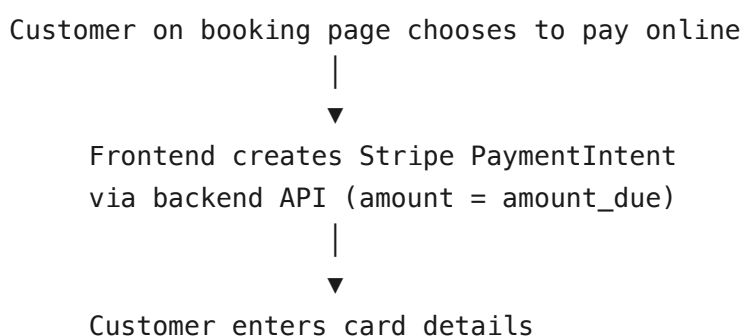
- Selling a product (skincare kit) with no appointment
  - Extra charge after the appointment
  - Custom package or bundle deal
  - Missed appointment / cancellation fee
- 

## 6.3 Record Payment (Manual)





## 6.4 Stripe Payment Flow (Online)





(Stripe Elements embedded in page)



Stripe processes the payment



Success

Failed



Stripe sends  
webhook:  
payment\_intent.  
succeeded

Stripe sends  
webhook:  
payment\_intent.  
payment\_failed



Backend:

Backend:

1. Create Payment  
(status:  
SUCCEEDED,  
gateway: STRIPE,  
gateway\_txn\_id:  
pi\_xxx)
  2. Update invoice  
amount\_paid,  
amount\_due,  
status
1. Create Payment  
(status:  
FAILED,  
gateway: STRIPE)
  2. No invoice  
update

**Important:** Never trust the frontend for payment confirmation. Always rely on Stripe webhooks to confirm payment status on the backend.

## 6.5 Refund Flow

Staff opens Invoice detail → Payments section



Clicks [Refund] next to a specific payment



Refund Modal:

- Refund amount

(max = original payment  
amount, can do partial)  
• Reason (optional)



System does:

1. Create Payment record:  
type: REFUND  
amount: negative  
linked\_payment\_id:  
original payment ID  
status: SUCCEEDED
2. Mark original payment  
status → REFUNDED
3. Update Invoice:  
amount\_paid -= refund  
amount\_due += refund
4. If Stripe payment:  
Trigger Stripe Refund  
API as well
5. Update invoice status:  
If fully refunded  
→ REFUNDED  
Else  
→ PARTIALLY\_PAID

## 6.6 Void / Cancel Invoice

Staff opens an unpaid Invoice



Clicks [Void Invoice]



Confirmation dialog:

"Are you sure? This cannot be undone."



System does:

1. Set invoice status  
→ VOID
2. No money movement
3. Invoice is now  
read-only

**Rule:** Can only void an invoice that has NO successful payments. If payments exist, they must be refunded first.

## 7. UI Locations

### 7.1 Portal Sidebar

Portal Sidebar:

- └ Dashboard
- └ Appointments
- └ Customers
- └ Invoices ← NEW TAB
- └ Services
- └ Doctors
- └ Settings

Only one new tab: **Invoices**. Payments are NOT a separate tab — they live inside the invoice detail.

### 7.2 Invoices List Page

Shows all invoices for the clinic with filters.

Invoices

[+ Create Invoice]

Status ▼

Date Range

Customer ▼

Search...

| #        | Customer    | Date   | Total   | Status  |
|----------|-------------|--------|---------|---------|
| INV-0042 | Sarah Ahmed | 20 Apr | £480.00 | PARTIAL |
| INV-0041 | James Ali   | 18 Apr | £300.00 | PAID    |
| INV-0040 | Noor Khan   | 15 Apr | £75.00  | OVERDUE |
| INV-0039 | Aisha Patel | 12 Apr | £600.00 | PAID    |

Filters:

- Status: All, Draft, Sent, Partially Paid, Paid, Overdue, Void, Refunded
- Date range: From / To date pickers
- Customer: Dropdown/search
- Search: By invoice number

7.3 Invoice Detail Page

Invoice #INV-2026-0042

Status: PARTIALLY PAID

[Void Invoice]

Customer

Sarah Ahmed

sarah@email.com

07xxx xxx xxx

Appointment

20 Apr 2026, 2:00pm

Dr Ahmed

[View Appointment]

Line Items

| Service            | Qty | Price   | Disc  | Total   |
|--------------------|-----|---------|-------|---------|
| Botox – Forehead   | 1   | £250.00 | £0.00 | £250.00 |
| Lip Filler – 0.5ml | 1   | £150.00 | £0.00 | £150.00 |
| Subtotal           |     |         |       | £400.00 |
| Discount           |     |         |       | £0.00   |

|            |         |
|------------|---------|
| VAT (20%)  | £80.00  |
| Total      | £480.00 |
| Paid       | £100.00 |
| Amount Due | £380.00 |

— Payments —

[+ Record Payment]

| Date   | Amount  | Method | Gateway | Status  |       |
|--------|---------|--------|---------|---------|-------|
| 16 Apr | £100.00 | Card   | Stripe  | Success | [Ref] |

Notes: First time client

### 7.4 Customer Tab — Balance Display

On the customer list, add an **Outstanding Balance** column:

Customers

| Name        | Phone | Outstanding Balance |
|-------------|-------|---------------------|
| Sarah Ahmed | 07xxx | £380.00             |
| James Ali   | 07xxx | £0.00               |
| Noor Khan   | 07xxx | £75.00              |

On the customer detail page, show a financial summary section:

Financial Summary

|                 |             |
|-----------------|-------------|
| Total Invoiced: | £1,200.00   |
| Total Paid:     | £800.00     |
| Outstanding:    | £400.00     |
| Last Payment:   | 12 Apr 2026 |

This is NOT a stored value. It is always calculated:

```
Outstanding = SUM(amount_due) from all invoices for this customer
```

### 7.5 Appointments Tab — Payment Status

On the appointment list, add a **Payment** column showing the linked invoice status:

| Appointments |             |           |               |         |  |
|--------------|-------------|-----------|---------------|---------|--|
| Date         | Customer    | Doctor    | Services      | Payment |  |
| 20 Apr       | Sarah Ahmed | Dr Ahmed  | Botox, Filler | Partial |  |
| 18 Apr       | James Ali   | Dr Fatima | Dermal Filler | Paid    |  |
| 15 Apr       | Noor Khan   | Dr Ahmed  | Consultation  | Unpaid  |  |

On the appointment detail page, show a payment summary section:

| Payment Summary                |  |
|--------------------------------|--|
| Invoice: #INV-2026-0042        |  |
| Total: £480.00                 |  |
| Paid: £100.00                  |  |
| Due: £380.00                   |  |
| Status: PARTIALLY PAID         |  |
| <a href="#">[View Invoice]</a> |  |

## 8. Discounts

### Line Item Discount

Applied to individual services on the invoice.

**Example:** 10% off Botox for a returning customer.

Botox – Forehead

Unit Price: £250.00  
Discount: 10% = £25.00  
Line Total: £225.00

## Invoice Level Discount

Applied to the entire invoice subtotal.

**Example:** £50 off total for a package deal.

Subtotal: £400.00  
Discount: £50.00 (fixed)  
After Discount: £350.00  
VAT (20%): £70.00  
Total: £420.00

## Discount Fields

Both invoice items and invoices have:

| Field           | Values              | Example          |
|-----------------|---------------------|------------------|
| discount_type   | PERCENTAGE or FIXED | PERCENTAGE       |
| discount_value  | The raw value       | 10 (meaning 10%) |
| discount_amount | Calculated in cents | 2500 (= £25.00)  |

Tax is calculated **after** discounts are applied.

## 9. Clinic Tax Settings

Each clinic configures their tax settings. These are stored in the clinic settings (not on every invoice — but they are **snapshotted** onto each invoice at creation time).

| Setting   | Example | Notes                 |
|-----------|---------|-----------------------|
| tax_rate  | 20      | Percentage (20 = 20%) |
| tax_label | "VAT"   | Displayed on invoices |

|               |       |                                     |
|---------------|-------|-------------------------------------|
| tax_inclusive | false | If true, prices already include tax |
|---------------|-------|-------------------------------------|

### Tax Inclusive vs Exclusive:

EXCLUSIVE (tax\_inclusive = false):

Service price: £100.00

Tax (20%): £20.00

Customer pays: £120.00

INCLUSIVE (tax\_inclusive = true):

Service price: £100.00 (already includes tax)

Tax portion: £16.67

Pre-tax: £83.33

Customer pays: £100.00

## 10. Invoice Numbering

Each clinic has its own sequential invoice numbers.

**Format:** INV-{YEAR}-{SEQUENCE}

### Examples:

- INV-2026-0001
- INV-2026-0002
- INV-2026-0042

### Rules:

- Sequence is per clinic — Clinic A and Clinic B both have their own INV-0001
- Numbers never repeat within a clinic
- Numbers are never reused (even if an invoice is voided)
- Reset yearly (optional clinic setting) — or continuous

## 11. Overdue Invoice Handling

A **background cron job** runs once per day (using BullMQ):

Every day at midnight:

|

▼

Find all invoices where:



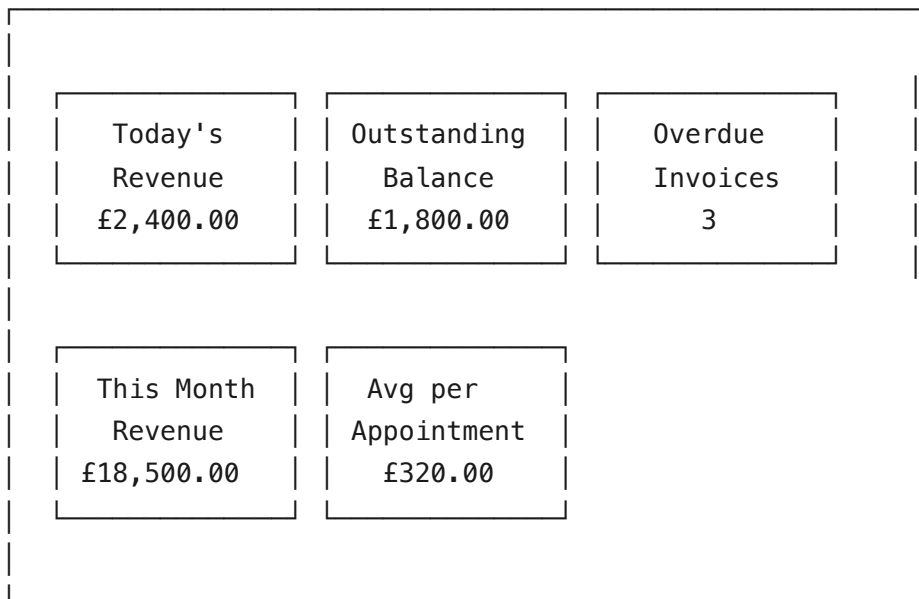
```
- status = SENT or PARTIALLY_PAID
- due_date < today
  |
  ▼
Set status → OVERDUE
  |
  ▼
(Future: send reminder email to customer)
```

This runs automatically. Staff doesn't need to manually mark invoices as overdue.

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## 12. Dashboard Stats

On the portal dashboard/home page, show financial summary cards:



All values are calculated, not stored:

- **Today's Revenue:** Sum of all payments with `paid_at = today` and `status = SUCCEEDED`
  - **Outstanding Balance:** Sum of `amount_due` from all non-VOID, non-PAID invoices
  - **Overdue Invoices:** Count of invoices with status = OVERDUE
  - **This Month Revenue:** Sum of payments this calendar month
  - **Avg per Appointment:** This month's revenue / number of completed appointments
-

## 13. Business Rules

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These are non-negotiable rules that must be followed throughout the implementation:

### Multi-Tenancy

- **Every single database query MUST filter by** `clinic_id` — this is the #1 review priority. No exceptions.

### Money

- All amounts stored as **integers in cents/pence** (25000 = £250.00)
- Never use floats or decimals for money
- Frontend converts for display: `(amount / 100).toFixed(2)`

### Balances

- **Never store balances separately** — always calculate from invoices and payments
- Customer outstanding = `SUM(amount_due)` from their invoices
- If performance becomes an issue later, we add caching — but not now

### Snapshots

- Invoice items snapshot the service `description` and `unit_price` at creation time
- Invoice snapshots the clinic's `tax_rate` and `tax_label` at creation time
- Changes to services or tax settings do NOT affect existing invoices

### Voiding

- Can only void invoices with zero successful payments
- If payments exist, refund them first, then void

### Refunds

- A refund creates a new Payment record with `type = REFUND` and negative amount
- The original payment's status changes to `REFUNDED`
- Stripe refunds must also trigger the Stripe Refund API

### Status Transitions

- Invoices move forward through statuses — never backwards (except via refund/void)
  - The `amount_paid` and `amount_due` fields drive status changes, not manual updates
-

## 14. Complete Example Walkthrough

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Let's follow a real scenario from start to finish.

### The Setup

- **Clinic:** Glow Aesthetics (clinic\_id: clinic\_abc)
  - **Customer:** Sarah Ahmed
  - **Services booked:** Botox - Forehead (£250), Lip Filler 0.5ml (£150)
  - **Doctor:** Dr Ahmed
  - **Date:** 20 April 2026, 2:00pm
  - **Clinic tax:** 20% VAT, exclusive
- 

### Step 1: Appointment Booked

Sarah books an appointment. The system creates:

**Appointment:**

- id: appt\_001
- status: CONFIRMED
- date: 20 Apr, 14:00–15:00

**Invoice (auto-generated):**

- id: inv\_001
- invoice\_number: INV-2026-0042
- subtotal: 40000 (£400.00)
- discount\_amount: 0
- tax\_rate: 20
- tax\_amount: 8000 (£80.00)
- total\_amount: 48000 (£480.00)
- amount\_paid: 0
- amount\_due: 48000
- status: DRAFT

**Invoice Items:**

| # | Description        | Qty | Unit Price | Discount | Line Total |
|---|--------------------|-----|------------|----------|------------|
| 1 | Botox - Forehead   | 1   | 25000      | 0        | 25000      |
| 2 | Lip Filler - 0.5ml | 1   | 15000      | 0        | 15000      |

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### Step 2: Staff Sends Invoice

Staff opens the invoice and clicks "Send" (marks as sent).

- Invoice status → **SENT**
- 

### Step 3: Sarah Pays £100 Deposit (Card via Stripe)

Stripe webhook fires `payment_intent.succeeded` .

#### Payment created:

- amount: 10000 (£100.00)
- type: PAYMENT
- payment\_method: CARD
- payment\_gateway: STRIPE
- gateway\_transaction\_id: pi\_abc123
- status: SUCCEEDED

#### Invoice updated:

- amount\_paid: 10000
  - amount\_due: 38000
  - status → **PARTIALLY\_PAID**
- 

### Step 4: Appointment Completed, Sarah Pays Remaining £380 (Cash)

Staff clicks [+ Record Payment]:

- Amount: £380.00
- Method: CASH
- Date: 20 Apr 2026

#### Payment created:

- amount: 38000 (£380.00)
- type: PAYMENT
- payment\_method: CASH
- payment\_gateway: MANUAL
- status: SUCCEEDED

#### Invoice updated:

- amount\_paid: 48000
  - amount\_due: 0
  - status → **PAID**
- 

### Final State

Invoice #INV-2026-0042

└─ Status: PAID

└─ Total: £480.00

└─ Paid: £480.00

└─ Due: £0.00

└─ Items:

└─ Botox - Forehead: £250.00

└─ Lip Filler - 0.5ml: £150.00

└─ Payments:

└─ 16 Apr - £100.00 - Card (Stripe) - Succeeded

└─ 20 Apr - £380.00 - Cash (Manual) - Succeeded

## 15. Priority Roadmap

| Priority | What                                                     | Why                                           |
|----------|----------------------------------------------------------|-----------------------------------------------|
| Now      | Invoice, InvoiceItems, Payments — DB schemas + full CRUD | Core foundation, nothing works without this   |
| Now      | Record payment flow (manual)                             | Clinics need to take and track money          |
| Next     | Discounts + tax settings                                 | Every clinic needs this from day one          |
| Next     | Refunds                                                  | Will come up fast once clinics start charging |
| Later    | Overdue cron job                                         | Automates status management                   |
| Later    | Dashboard stats                                          | After real data is flowing                    |
| Later    | Stripe integration                                       | When we add online booking payments           |

**Not included in this phase:** Invoice PDF generation and email sending — will be a separate ticket.

## 16. Enums Reference

For the developer to define in `packages/shared-types/` :

**InvoiceStatus:** DRAFT | SENT | PARTIALLY\_PAID | PAID | OVERDUE | VOID |  
REFUNDED

**PaymentType:** PAYMENT | REFUND

**PaymentMethod:** CASH | CARD | BANK\_TRANSFER

**PaymentGateway:** MANUAL | STRIPE

**PaymentStatus:** SUCCEEDED | FAILED | REFUNDED

**DiscountType:** PERCENTAGE | FIXED

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*End of document.*